Decide whether retrieval is required for your agent. Clearly explain your reasoning.

- My travel agent would utilize both the traditional retrieval approach and MMR. It would require traditional for more specific queries such as what time is my flight, and find flight details on this day for Delta airline. These would require the most closely related data. Other types of queries that are more general, such as plan me a day trip for NYC, would benefit from a more broad coverage search.

If retrieval is required, describe how you will integrate a semantic retrieval mechanism using an external data source.

- For semantic retrieval, I would call api calls (e.g., Tavily for search api), visit blog posts (e.g., Reddit), and view other established webpages (e.g., official park pages) to get attractions and other relevant details.

Demonstrate that retrieval meaningfully influences the agent's output (e.g., by explaining an example in which retrieved context changes or grounds the response).

- A retrieval mechanism is critical to the entire agentic process. If I were to ask the user to find me the cheapest prices for a ticket to Seattle, WA for September 20-25, 2026, without retrieval, it would either hallucinate, provide me wrong answers, or inform me that it's unable to. With retrieval implemented, it would be able to connect to the web, find the best matches, and inform me of them.

Briefly describe key retrieval design choices, such as data source selection, document segmentation, or chunking approaches, and how many results are retrieved.

- I would use Duffel as the main API source that would provide access to flights, stays, and car rentals. These wouldn't be stored in any databases or be chunked because these are typically outputted in JSON format. For results, depending on whether it's precise or exploratory, it would fetch 10 or 25 results, respectively. With each one, it would then apply reranking and other filters to narrow the results to around 5 final outputs.
- It would also scrape the web, such as reddit or other travel blogs, to learn more about tourist attractions. These would most likely just do standard size chunks of roughly 500 tokens with an overlap of 100 tokens. Adaptive chunking wouldn't be required as most of these posts are written in prose with no real structure or formats.

Identify one potential retrieval-related failure mode and explain how your design reduces or manages that risk.

- One risk is that when asked for an exploratory request to create a trip to anywhere, the retrieval might be too narrow. To combat this, the agent would have an adaptive retrieval system that broadens the MMR search based on the query. If the LLM recognizes it as broad/exploratory, it would categorize the prompt as such. There would be two categories, specific or exploratory.

Exploratory would focus on MMR search with a parameter to specify the lambda value. Specific would focus on traditional retrieval.

If you determine that retrieval is not required for your agent, provide a clear architectural justification for why a non-RAG design remains reliable for your use case.

- NA